5 Tensor Analysis

5.1 General Rule of Transformation of Coordinate Differentials

Consider two reference frames (RF) Σ and Σ'. If the coordinates $\{x'^\mu\} = (x'^0, x'^1, x'^2, x'^3)$ in Σ' be related to the coordinates $\{x^\mu\} = (x^0, x^1, x^2, x^3)$ in Σ by a coordinate transformation:

$$x'^\mu = x'^\mu\left(x^0, x^1, x^2, x^3\right), \qquad \mu = 0, 1, 2, 3, \tag{5.1}$$

then the necessary and sufficient condition that the four components of x'^μ form a set of independent, real functions of $\left(x^0, x^1, x^2, x^3\right)$ is that their *Jacobian*:

$$\mathcal{J}(x \to x') = \left|\frac{\partial x'}{\partial x}\right| = \begin{vmatrix} \frac{\partial x'^0}{\partial x^0} & \cdots & \frac{\partial x'^3}{\partial x^0} \\ \vdots & & \vdots \\ \frac{\partial x'^0}{\partial x^3} & \cdots & \frac{\partial x'^3}{\partial x^3} \end{vmatrix} \neq 0. \tag{5.2}$$

In such case, one can have the inverse coordinate transformations as:

$$x^\mu = x^\mu\left(x'^0, x'^1, x'^2, x'^3\right), \qquad \mu = 0, 1, 2, 3. \tag{5.3}$$

At any point in a general spacetime, the direction (of a vector) in a coordinate frame, say Σ, is determined by the coordinate differentials dx^μ defined in that frame. In another frame, say Σ', the direction is determined by the associated coordinate differentials dx'^ν. The relation between the coordinate differentials in the two frames is given by the transformation rules:

$$dx'^\mu = \frac{\partial x'^\mu}{\partial x^\nu}\, dx^\nu, \quad \text{and} \quad dx^\mu = \frac{\partial x^\mu}{\partial x'^\nu}\, dx'^\nu. \tag{5.4}$$

While carrying out such a transformation, we have

$$\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\alpha}{\partial x'^\nu} = \delta^\mu{}_\nu, \quad \text{and} \quad \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x'^\mu}{\partial x^\beta} = \delta^\alpha{}_\beta. \tag{5.5}$$

So, if one considers $\partial x'^\mu / \partial x^\alpha$ as a matrix — the *transformation matrix*, for transformation from the x^α coordinate system to the x'^α coordinate system — then $\partial x^\alpha / \partial x'^\mu$ is its inverse, i.e., the transformation matrix for the inverse transformation from the x'^α coordinate system to the x^α coordinate system. Consequently, the Jocobians of the transformation and the inverse transformation satisfy:

$$\mathcal{J}(x \to x')\mathcal{J}(x' \to x) = \left|\frac{\partial x'}{\partial x}\right|\left|\frac{\partial x}{\partial x'}\right| = 1. \tag{5.6}$$

In Special Relativity (SR), the transformation matrices are the LT matrices, and therefore the Jacobian is identically equal to unity:

$$\frac{\partial x'^\mu}{\partial x^\nu} \stackrel{\text{SR}}{=} \Lambda^\mu{}_\nu, \qquad \mathcal{J}(x \to x') = \left|\frac{\partial x'}{\partial x}\right| \stackrel{\text{SR}}{=} \det\left(\Lambda^\mu{}_\nu\right) = 1. \tag{5.7}$$

5.2 Definition and Categorization of tensors

Tensors (of specific *rank* and *type*) are physical entities associated with a spacetime point, which follow linear, homogeneous laws of transformation under the transformation of coordinates. A tensor carries superscript and subscript indices, the total number of which signifies the so-called *rank* of the tensor:

$$T^{\alpha_1\alpha_2\ldots\alpha_m}{}_{\beta_1\beta_2\ldots\beta_n}(x^\mu) \qquad \Rightarrow \quad \text{Rank} = (m+n). \tag{5.8}$$

In general, each component of a tensor is a function of the coordinates x^μ $(\mu = 0, 1, 2, 3)$.

5.2.1 Tensor of Rank 0 : Scalar

Rank 0 tensors are scalars which remain invariant under the transformation of coordinates:

$$\phi\,(x^\mu) \quad \xrightarrow{\text{Inv.}} \text{ Coordinate transformations } x^\mu \to x'^\mu\,. \tag{5.9}$$

In SR: a (space-time) scalar, invariant under the Lorentz transformations, is often called a *Lorentz scalar* or a *Four-scalar*, e.g., the invariant distance squared $s^2 = c^2t^2 - |\vec{\boldsymbol{x}}|^2$.

5.2.2 Tensor of Rank 1 : Vector

Rank 1 tensors are vectors, which are of two types:

I. Contravariant vector: denoted by

$$A^\alpha(x^\mu) \;=\; \Big(A^0(x^\mu), A^1(x^\mu), A^2(x^\mu), A^3(x^\mu)\Big)\,,$$

transforms like the coordinate differentials dx^α, under a given coordinate transformation:

$$\boxed{A'^\mu \;=\; \frac{\partial x'^\mu}{\partial x^\alpha}\, A^\alpha} \quad \underset{\text{to}}{\xrightarrow{\text{similar}}} \quad dx'^\mu \;=\; \frac{\partial x'^\mu}{\partial x^\alpha}\, dx^\alpha\,. \tag{5.10}$$

In SR: the above transformation rule becomes

$$\boxed{A'^\mu \;=\; \Lambda^\mu{}_\alpha\, A^\alpha}\,, \tag{5.11}$$

and a contravariant vector is often called a (space-time) *Four-vector*.

II. Covariant vector: denoted by

$$B_\beta(x^\mu) \;=\; \Big(B_0(x^\mu), B_1(x^\mu), B_2(x^\mu), B_3(x^\mu)\Big)\,,$$

transforms like the partial derivatives $\partial/\partial x^\beta$ of a scalar $\phi(x^\mu)$, under a given coordinate transformation:

$$\boxed{B'_\nu \;=\; \frac{\partial x^\beta}{\partial x'^\nu}\, B_\beta} \quad \underset{\text{to}}{\xrightarrow{\text{similar}}} \quad \frac{\partial \phi}{\partial x'^\nu} \;=\; \frac{\partial x^\beta}{\partial x'^\nu}\,\frac{\partial \phi}{\partial x^\beta}\,. \tag{5.12}$$

In SR: the above transformation rule becomes

$$\boxed{B'_\nu \;=\; \Lambda_\nu{}^\beta\, B_\beta}\,, \tag{5.13}$$

and a covariant vector is often called a (space-time) *co-vector*.

One may also note, by analogy with the ordinary three-dimensional Euclidean space, that the partial derivatives of a scalar ϕ w.r.to the coordinates x^μ constitute the gradient of the scalar, which points along the normal to the surface $\phi =$ constant. Accordingly, the components of a covariant vector B_β may be thought of as the *normal*, or *perpendicular*, projections of the vector on the coordinate axes. By a similar reasoning, the components of a contravariant vector A^α may be thought of as the *tangential*, or *parallel*, projections of the vector on the coordinate axes. Our familier perception is that the parallel projections define the components of a vector in the Euclidean three-dimensional space. Hence, it is convenient to consider the contravariant vectors to form a vector space in the usual sense, whereas the covariant vectors constitute a *dual* vector space. Such a duality is of course the duality of projections in the coordinate space, and not to be confused with the *tensor duality* which we shall discuss later on.

5.2.3 Tensors of Rank ≥ 2

These are of three types:

I. Contravariant tensor: of say rank $m\,(\geq 2)$, denoted by $T^{\alpha_1\alpha_2\ldots\alpha_m}(x^\mu)$, transforms like a direct product of m contravariant vectors, under a given coordinate transformation:

$$\boxed{T'^{\mu_1\ldots\mu_m} \;=\; \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}}\cdots\frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}}\;T^{\alpha_1\ldots\alpha_m}} \quad \xrightarrow[\text{to}]{\text{similar}} \; A'^{\mu_1}\ldots A'^{\mu_m} \;=\; \frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}}\cdots\frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}}\;A^{\alpha_1}\ldots A^{\alpha_m}\,. \tag{5.14}$$

In SR: the above transformation rule becomes

$$\boxed{T'^{\mu_1\ldots\mu_m} \;=\; \Lambda^{\mu_1}_{\ \ \alpha_1}\ldots\Lambda^{\mu_m}_{\ \ \alpha_m}\;T^{\alpha_1\ldots\alpha_m}}\,. \tag{5.15}$$

II. Covariant tensor: of say rank $n\,(\geq 2)$, denoted by $T_{\beta_1\beta_2\ldots\beta_n}(x^\mu)$, transforms like a direct product of n covariant vectors, under a given coordinate transformation:

$$\boxed{T'_{\nu_1\ldots\nu_n} \;=\; \frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}}\cdots\frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}}\;T_{\beta_1\ldots\beta_n}} \quad \xrightarrow[\text{to}]{\text{similar}} \; B'_{\nu_1}\ldots B'_{\nu_n} \;=\; \frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}}\cdots\frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}}\;B_{\beta_1}\ldots B_{\beta_n}\,. \tag{5.16}$$

In SR: the above transformation rule becomes

$$\boxed{T'_{\nu_1\ldots\nu_n} \;=\; \Lambda_{\nu_1}^{\ \ \beta_1}\ldots\Lambda_{\nu_n}^{\ \ \beta_n}\;T_{\beta_1\ldots\beta_n}}\,. \tag{5.17}$$

III. Mixed tensor: of say rank $(m+n)\,(\geq 2,\, m\geq 1, n\geq 1)$, denoted by $T^{\alpha_1\alpha_2\ldots\alpha_m}{}_{\beta_1\beta_2\ldots\beta_n}(x^\mu)$, transforms like a direct product of m contravariant and n covariant vectors, under a given coordinate transformation:

$$\boxed{T'^{\mu_1\ldots\mu_m}{}_{\nu_1\ldots\nu_n} \;=\; \left(\frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}}\cdots\frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}}\right)\left(\frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}}\cdots\frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}}\right)T^{\alpha_1\ldots\alpha_m}{}_{\beta_1\ldots\beta_n}} \tag{5.18}$$

$$\xrightarrow[\text{to}]{\text{similar}} \left(A'^{\mu_1}\ldots A'^{\mu_m}\right)\left(B'_{\nu_1}\ldots B'_{\nu_n}\right) = \left(\frac{\partial x'^{\mu_1}}{\partial x^{\alpha_1}}\cdots\frac{\partial x'^{\mu_m}}{\partial x^{\alpha_m}}\right)\left(\frac{\partial x^{\beta_1}}{\partial x'^{\nu_1}}\cdots\frac{\partial x^{\beta_n}}{\partial x'^{\nu_n}}\right)\left(A^{\alpha_1}\ldots A^{\alpha_m}\right)\left(B_{\beta_1}\ldots B_{\beta_n}\right).$$

In SR: the above transformation rule becomes

$$\boxed{T'^{\mu_1\ldots\mu_m}{}_{\nu_1\ldots\nu_n} \;=\left(\Lambda^{\mu_1}_{\ \ \alpha_1}\ldots\Lambda^{\mu_m}_{\ \ \alpha_m}\right)\left(\Lambda_{\nu_1}^{\ \ \beta_1}\ldots\Lambda_{\nu_n}^{\ \ \beta_n}\right)T^{\alpha_1\ldots\alpha_m}{}_{\beta_1\ldots\beta_n}}\,. \tag{5.19}$$

5.3 Tensor Algebra

5.3.1 Linear Combination of Tensors

Only tensors of the same rank and type (contravariant, covariant or mixed) could be added to produce another tensor of identical rank and type. The same is true, in fact, for any linear combination of tensors of same rank and type. For e.g. given two covariant tensors of rank 2, $A_{\alpha\beta}$ and $B_{\alpha\beta}$, the linear combination

$$T_{\alpha\beta} \;=\; \phi\;A_{\alpha\beta} \;+\; \xi\;B_{\alpha\beta}\;, \qquad [\text{where } \phi,\xi \text{ are space-time scalars}]\,, \tag{5.20}$$

also transforms like a covariant tensor of rank 2.

5.3.2 Multiplication of Tensors

Unlike the linear combination, the multiplication of tensors of any type and rank, same or different, is possible. Accordingly, one can define three kinds of tensor products:

I. Direct Product: The aggregate obtained via multiplication of any two tensors of the same or different type, and of ranks m and n say, is another tensor of rank $(m+n)$. For e.g.

$$A^\alpha\, B_\beta \;=\; T^\alpha_{\ \beta}\;,\quad A^{\alpha\beta}\, B^\gamma \;=\; T^{\alpha\beta\gamma}\;,\quad A^\alpha\, B_{\beta\gamma}{}^\delta \;=\; T^\alpha{}_{\beta\gamma}{}^\delta\;,\quad \cdots\,. \tag{5.21}$$

II. Contracted Product: If two different types of tensors of ranks m and n say, have p number of matching contra and covariant indices, then the direct product of the tensors reduces to another tensor of rank $(m+n-2p)$ with no matching contra and covariant indices. For e.g.

$$A^{\alpha\beta}\, B_{\beta\gamma} \;=\; T^\alpha{}_\gamma\;,\quad A^{\alpha\beta}{}_{\mu\nu}\, B_{\gamma\delta}{}^{\mu\nu} \;=\; T^{\alpha\beta}{}_{\gamma\delta}\;,\quad A^{\alpha\beta}{}_\gamma\, B_\beta{}^\gamma \;=\; T^\alpha\;,\quad \cdots\,. \tag{5.22}$$

Contraction implies the reduction of any mixed tensor of rank n (>1), and with p matching contra and covariant indices, to a new tensor of rank $(n-2p)$ after summing over those p indices. For e.g.

$$T^\mu{}_{\alpha\mu} \;=\; T_\alpha\;,\quad T^\mu{}_{\alpha\mu\beta} \;=\; T_{\alpha\beta}\;,\quad T^{\mu\alpha}{}_\mu{}^{\nu\beta}{}_{\nu\gamma} \;=\; T^{\alpha\beta}{}_\gamma\;,\quad \cdots\,. \tag{5.23}$$

III. Scalar Product: Scalar function which is the outcome of a fully contracted product of two tensors of the same rank and when the contra(co)variant indices of one is identical to the co(contra)variant indices of the other. For e.g.

$$A\cdot B \;:=\; A^\alpha\, B_\alpha\;,\qquad S\cdot T \;:=\; S^{\alpha\beta}\, T_{\alpha\beta}\;,\qquad Q\cdot R \;:=\; Q^{\alpha\beta}{}_\gamma\, R_{\alpha\beta}{}^\gamma\;,\quad \cdots\,. \tag{5.24}$$

5.3.3 Norm and Trace of Tensors

The *norm* of a tensor of any rank is its scalar product with itself. For e.g.

$$\parallel A\parallel^2 := A_\alpha\, A^\alpha\;,\qquad \parallel S\parallel^2 := S_{\alpha\beta}\, S^{\alpha\beta}\;,\qquad \parallel T\parallel^2 := T_{\alpha\beta\gamma}\, T^{\alpha\beta\gamma}\;,\quad \cdots\,. \tag{5.25}$$

Unlike the norm, the *trace* is defined only for tensors of even rank (≥ 2). The trace is the full contraction of the indices of the even rank tensor in the mixed form. For e.g.

$$\mathrm{tr}\left(S^{\alpha\beta}\right) \;:=\; S_\alpha{}^\alpha\;,\qquad \mathrm{tr}\left(T^{\alpha\beta\gamma\delta}\right) \;:=\; T_\alpha{}^\alpha{}_\beta{}^\beta\;,\quad \cdots\,. \tag{5.26}$$

5.4 The Metric Tensor

5.4.1 Definition and Properties

Let us recall the foremost criterion in the formulation of a relativity theory, that the corresponding transformation equations must leave invariant the line element ds^2, i.e. the squared separation between two infinitesimally close points in space, or events in space-time. For e.g. in Galilean relativity, the corresponding Galilean transformation equations leave invariant the infinitesimal squared separation of points in the three-dimensional Euclidean space:

$$ds^2 \;\overset{\text{Euc}}{\equiv}\; |d\vec{\boldsymbol{x}}|^2 \;=\left(dx^1\right)^2+\left(dx^2\right)^2+\left(dx^3\right)^2\;, \tag{5.27}$$

whereas in special relativity, the corresponding Lorentz or Poincaŕe transformation equations leave invariant the infinitesimal squared separation of events in the four-dimensional Minkowski space-time:

$$ds^2 \;\overset{\text{SR}}{\equiv}\; (dx^0)^2 - \;(dx^1)^2 - \;(dx^2)^2 - \;(dx^3)^2\;. \tag{5.28}$$

Since ds^2 is quadratic in the differentials of the coordinates (dx^α , where $\alpha = 0, 1, 2, 3$ in general), one may express it (as any other quadratic):

$$\boxed{ds^2 \;=\; g_{\alpha\beta}\, dx^\alpha\, dx^\beta}\,, \tag{5.29}$$

where the coefficients $g_{\alpha\beta}$ $(\alpha, \beta = 0, 1, 2, 3)$ form the elements of a covariant tensor of rank 2, known as the *metric tensor*, or sometimes simply the *metric*. In general, $g_{\alpha\beta}$ could be a function of the coordinates, and not necessarily diagonal, i.e. there could be cross terms in ds^2, such as dx^1dx^2, dx^2dx^3, $dx^0dx^1, \cdots$ (unlike in the Euclidean space or the Minkowski space-time, with ds^2 given by Eqs. (5.27) and (5.28) respectively).

That the metric $g_{\alpha\beta}$ indeed transforms as a covariant tensor of rank 2, can be verified at once: if, let's say, the above relation (5.29) holds in one frame of reference Σ, then in another frame Σ', characterized by coordinates x'^μ, the corresponding relation is

$$ds'^2 \;=\; g'_{\mu\nu}\, dx'^\mu\, dx'^\nu \;. \tag{5.30}$$

Since $ds'^2 = ds^2$ under the transformation $x^\alpha \longrightarrow x'^\alpha$, we have

$$g'_{\mu\nu}\, dx'^\mu\, dx'^\nu \;=\; g_{\alpha\beta}\, dx^\alpha\, dx^\beta \qquad \Longrightarrow \qquad \underline{\underline{g'_{\mu\nu} \;=\; \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\, g_{\alpha\beta}}}\;, \qquad [\text{QED}]\,. \tag{5.31}$$

The following are the properties of the metric tensor:

(i) It is *symmetric* under the interchange of indices, i.e.

$$\boxed{g_{\alpha\beta} \;=\; g_{\beta\alpha}}\,, \qquad \forall \alpha, \beta\;. \tag{5.32}$$

(ii) Its contracted product with the metric tensor in contravariant form, gives the Kronecker delta, which can be shown to transform as a mixed tensor of rank 2:

$$\boxed{g_{\mu\alpha}\, g^{\mu\beta} \;=\; \delta_\alpha^{\;\beta}}\,, \qquad \delta_\alpha^{\;\beta} = \begin{cases} 1 & \text{for } \alpha = \beta \\ 0 & \text{otherwise} \end{cases}\;. \tag{5.33}$$

The symmetry property (5.32) ensures there are at most $4\,(4+1)\,/2 = 10$ independent components of $g_{\alpha\beta}$ in a $(3+1)$-dimensional space-time. Accordingly, one can decompose the r.h.s. of Eq. (5.29) as[¶] :

$$ds^2 \;=\; \underset{\left[\begin{array}{c}\text{ONE term for}\\ \mu=\nu=0.\end{array}\right]}{g_{00}\,(dx^0)^2} \;+\; \underset{\left[\begin{array}{l}\text{Sum of THREE terms for}\\ \mu=0,\nu=i.\ \text{The factor of}\\ 2\text{ is for symmetry } g_{0i}=g_{i0}.\end{array}\right]}{2g_{0i}\; dx^0\; dx^i} \;+\; \underset{\left[\begin{array}{l}\text{Sum of SIX terms for}\\ \mu=i,\nu=j,\ \text{due to}\\ \text{symmetry } \ g_{ij}=g_{ji}\,.\end{array}\right]}{g_{ij}\; dx^i\; dx^j} \;. \tag{5.34}$$

The property (5.33), on the other hand, implies that the norm of the metric tensor $g_{\alpha\beta}$ is

$$\parallel g \parallel^2 \;=\; \delta_\alpha^{\;\alpha} \;=\; 4 \quad \text{in } (3+1)\text{-dimensions.} \tag{5.35}$$

5.4.2 Usage: Raising and Lowering of Tensor indices

Raising and lowering of tensor indices are done by contracting with the metric tensor $g_{\alpha\beta}$. For e.g.

$$\begin{aligned} A_\alpha \;&=\; g_{\alpha\beta}\, A^\beta\;, \quad A^\alpha \;=\; g^{\alpha\beta}\, A_\beta\;; \\ T_{\mu\nu} \;&=\; g_{\mu\alpha}\, g_{\nu\beta}\, T^{\alpha\beta}\;, \quad T^{\mu\nu} \;=\; g^{\mu\alpha}\, g^{\nu\beta}\, T_{\alpha\beta}\;; \\ T_{\mu\nu\lambda\ldots}^{\quad\;\;\alpha\beta\gamma\ldots} \;&=\; \left(g^{\alpha\rho}\, g^{\beta\sigma}\, g^{\gamma\tau}\ldots\right) T_{\mu\nu\lambda\ldots\rho\sigma\tau\ldots}\;, \quad \cdots\;. \end{aligned} \tag{5.36}$$

[¶]It is to be noted that in this section, and throughout, the Greek indices $(\alpha, \beta, \gamma, \ldots)$ run from 0 to 3, i.e. over the entire $(3+1)$-dimensional space-time, whereas the Latin indices $(i, j, k, \ldots)$ run from 1 to 3, i.e. over the 3-dimensional spatial hypersurface.

Accordingly, the scalar products are expressed as

$$\begin{aligned} A \cdot B &:= A^{\alpha} B_{\alpha} = g^{\alpha\beta} A_{\beta} B_{\alpha} = A_{\beta} B^{\beta} \equiv A_{\alpha} B^{\alpha} , \\ S \cdot T &:= S^{\alpha\beta} T_{\alpha\beta} = g^{\mu\alpha} g^{\nu\beta} S_{\mu\nu} T_{\alpha\beta} = S_{\mu\nu} T^{\mu\nu} \equiv S_{\alpha\beta} T^{\alpha\beta} , \quad \cdots . \end{aligned} \tag{5.37}$$

One may also note that for the metric tensor itself the trace is equal to the norm:

$$g_{\alpha\beta} g^{\alpha\beta} = g_{\alpha}^{\ \alpha} = \delta_{\alpha}^{\ \alpha} = \| g \|^2 . \tag{5.38}$$

5.4.3 The Minkowski Metric

Special Relativity (SR) alludes to a *special* form of the metric tensor: $g_{\alpha\beta} \equiv \eta_{\alpha\beta}$, referred to as the *Minkowski metric*, which is diagonal, and with each non-vanishing element normalized to unity in magnitude:

$$\eta_{\alpha\beta} = \mathrm{diag}\,(1, -1, -1, -1) \qquad \text{i.e.,} \qquad |\eta_{\alpha\beta}| = \delta_{\alpha\beta} . \tag{5.39}$$

Explicitly:

$$\eta_{00} = 1 \quad , \qquad \eta_{0i} = 0 \quad (\forall\ i = 1,2,3) \quad , \qquad \eta_{ij} = -\delta_{ij} \quad (\forall\ i,j = 1,2,3) , \tag{5.40}$$

and one may at once verify that the line element in Special Relativity is thus given above by Eq. (5.28).

5.4.4 Vectors and Tensors in Minkowski space-time

Let us now see how the Minkowski metric tensor $\eta_{\alpha\beta}$ helps in identifying or(and) correlating the contra and covariant components of a vector or a tensor in special relativity.

Components of a Vector: The components of a covariant vector A_α are related to the components of the corresponding contravariant vector A^α via

$$A_{\alpha} = \eta_{\alpha\beta} A^{\beta} \qquad \Longrightarrow \qquad \begin{cases} A_0 = \eta_{0\beta} A^{\beta} = \eta_{00} A^0 = A^0 , \\ A_i = \eta_{i\beta} A^{\beta} = \eta_{ij} A^j = -\delta_{ij} A^j . \end{cases} \tag{5.41}$$

Since in the ordinary Euclidean space, the parallel projections A^i define the components of a (three-)vector $\vec{\boldsymbol{A}}$, we have

$$A^{\alpha} := \left(A^0, A^i\right) \equiv \left(A^0, \vec{\boldsymbol{A}}\right) \qquad \text{and} \qquad A_{\alpha} := (A_0, A_i) = \left(A^0, -\delta_{ij} A^j\right) \equiv \left(A^0, -\vec{\boldsymbol{A}}\right) . \tag{5.42}$$

Components of a second rank Tensor: The covariant and contravariant components are related as

$$T_{\alpha\beta} = \eta_{\mu\alpha} \eta_{\nu\beta} T^{\mu\nu} \qquad \Longrightarrow \qquad \begin{cases} T_{00} = \eta_{0\mu} \eta_{0\nu} T^{\mu\nu} = (\eta_{00})^2 T^{00} = T^{00} , \\ T_{0i} = \eta_{0\mu} \eta_{i\nu} T^{\mu\nu} = \eta_{00} \eta_{ij} T^{0j} = -\delta_{ij} T^{0j} , \\ T_{i0} = \eta_{i\mu} \eta_{0\nu} T^{\mu\nu} = \eta_{ij} \eta_{00} T^{j0} = -\delta_{ij} T^{j0} , \\ T_{ij} = \eta_{i\mu} \eta_{j\nu} T^{\mu\nu} = \eta_{ik} \eta_{jl} T^{kl} = \delta_{ik} \delta_{jl} T^{kl} . \end{cases} \tag{5.43}$$

Corollary: If we choose, $T_{\alpha\beta} \equiv \eta_{\alpha\beta}$, then for the Minkowski metric tensor $\eta_{\alpha\beta}$ itself:

$$\eta_{00} = \eta^{00} = 1 , \quad \eta_{0i} = 0 = \eta^{0i} , \quad \eta_{ij} = -\delta_{ij} \iff \eta^{ij} = -\delta^{ij} . \tag{5.44}$$

This implies that $\eta_{\alpha\beta}$ and $\eta^{\alpha\beta}$ have the same set of components, viz. $\mathrm{diag}\,(1, -1, -1, -1)$.

For higher rank tensors, the relationship between the contra and covariant components could be found in a similar manner as above.

5.5 Tensor Symmetry and Antisymmetry

5.5.1 Symmetric and Antisymmetric Tensors

- Symmetry or antisymmetry is defined only for contravariant and covariant tensors of rank ≥ 2.
- Symmetry or antisymmetry could either be (i) *partial*, i.e. under the exchange of only a given pair of indices, or (ii) *complete*, i.e. under the exchange of any pair of indices.
- Symmetry or antisymmetry of some pattern (partial or complete) of a covariant tensor implies symmetry or antisymmetry of the same pattern of the corresponding contravariant tensor, and vice versa.

Let us illustrate these as follows:

Rank 2 tensors:

$$\text{Symmetric:} \qquad S_{\alpha\beta} = S_{\beta\alpha} \iff S^{\alpha\beta} = S^{\beta\alpha}\,, \tag{5.45}$$

$$\text{Antisymmetric:} \qquad A_{\alpha\beta} = -A_{\beta\alpha} \iff A^{\alpha\beta} = -A^{\beta\alpha} \implies A_{\alpha\alpha} = A^{\alpha\alpha} = 0\,. \tag{5.46}$$

Rank 3 tensors:

$$\text{Symmetric:} \qquad \begin{cases} \text{Partial:} \quad S_{\alpha\beta\gamma} = S_{\beta\alpha\gamma}\,, \text{ or } \quad S_{\alpha\beta\gamma} = S_{\alpha\gamma\beta}\,, \text{ or } \quad S_{\alpha\beta\gamma} = S_{\gamma\beta\alpha}\,; \\ \text{Complete:} \quad S_{\alpha\beta\gamma} = S_{\beta\gamma\alpha} = S_{\gamma\alpha\beta} = S_{\beta\alpha\gamma} = S_{\alpha\gamma\beta} = S_{\gamma\beta\alpha}\,, \end{cases} \tag{5.47}$$

$$\text{Antisymmetric:} \qquad \begin{cases} \text{Partial:} \quad A_{\alpha\beta\gamma} = -A_{\beta\alpha\gamma}\,, \text{ or } \quad A_{\alpha\beta\gamma} = -A_{\alpha\gamma\beta}\,, \text{ or } \quad A_{\alpha\beta\gamma} = -A_{\gamma\beta\alpha}\,; \\ \text{Complete:} \quad A_{\alpha\beta\gamma} = A_{\beta\gamma\alpha} = A_{\gamma\alpha\beta} = -A_{\beta\alpha\gamma} = -A_{\alpha\gamma\beta} = -A_{\gamma\beta\alpha}\,. \end{cases} \tag{5.48}$$

Similarly, one can have partial or complete symmetry or antisymmetry of higher rank tensors.

5.5.2 Symmetrization and Antisymmetrization of Tensors

Rank 2 tensors: Any (in general asymmetric, and let's say covariant) tensor $T_{\alpha\beta}$ could be split up as

$$\boxed{T_{\alpha\beta} = T_{(\alpha\beta)} + T_{[\alpha\beta]}}\,, \tag{5.49}$$

where

$$T_{(\alpha\beta)} = T_{(\beta\alpha)} = \frac{1}{2}\left(T_{\alpha\beta} + T_{\beta\alpha}\right)\,, \qquad T_{[\alpha\beta]} = -T_{(\beta\alpha)} = \frac{1}{2}\left(T_{\alpha\beta} - T_{\beta\alpha}\right)\,. \tag{5.50}$$

The symmetric part $T_{(\alpha\beta)}$ is called the *symmetrization* of $T_{\alpha\beta}$, whereas the antisymmetric part $T_{[\alpha\beta]}$ is called the *antisymmetrization* of $T_{\alpha\beta}$. Both these parts transform like second rank covariant tensors. Such a splitting is also possible for a second rank contravariant tensor $T^{\alpha\beta}$. However, a splitting into a (completely) symmetrized part and a (completely) antisymmetrized part is not possible for tensors of rank > 2.

Rank 3 tensors: Symmetrization and antisymmetrization of, let's say, a covariant tensor $T_{\alpha\beta\gamma}$ give

$$\begin{aligned} T_{(\alpha\beta\gamma)} &= \frac{1}{3!}\left(T_{\alpha\beta\gamma} + T_{\beta\gamma\alpha} + T_{\gamma\alpha\beta} + T_{\beta\alpha\gamma} + T_{\alpha\gamma\beta} + T_{\gamma\beta\alpha}\right)\,, \\ T_{[\alpha\beta\gamma]} &= \frac{1}{3!}\left(T_{\alpha\beta\gamma} + T_{\beta\gamma\alpha} + T_{\gamma\alpha\beta} - T_{\beta\alpha\gamma} - T_{\alpha\gamma\beta} - T_{\gamma\beta\alpha}\right)\,. \end{aligned} \tag{5.51}$$

By definition, $T_{(\alpha\beta\gamma)}$ and $T_{[\alpha\beta\gamma]}$ are completely symmetric and completely antisymmetric, respectively. One may also note that $T_{[\alpha\beta\gamma]}$ has non-vanishing components only for $\alpha \neq \beta \neq \gamma$. However, as mentioned above, addition (or any linear combination) of $T_{(\alpha\beta\gamma)}$ and $T_{[\alpha\beta\gamma]}$ does not give the original tensor $T_{\alpha\beta\gamma}$.

In principle, symmetrization and antisymmetrization of any general, let's say, covariant of rank $p\,(\geq 2)$, viz. $T_{\alpha_1\alpha_2\ldots\alpha_p}$, in a D-dimensional spacetime (i.e. 1 time + $(D-1)$ space), give

$$T_{(\alpha_1\alpha_2\ldots\alpha_p)} = \frac{1}{p!}\sum_P T_{\alpha_{P(1)}\alpha_{P(2)}\ldots\alpha_{P(p)}}\,, \tag{5.52}$$

$$T_{[\alpha_1\alpha_2\ldots\alpha_p]} = \frac{1}{p!}\sum_P (-1)^P\, T_{\alpha_{P(1)}\alpha_{P(2)}\ldots\alpha_{P(p)}}\,, \tag{5.53}$$

where the summations are for all permutations P of the indices $(\alpha_1\alpha_2\ldots\alpha_p)$ and the factor $(-1)^P$ indicates the parity — equal to -1 of odd permutations P, and $+1$ for even permutations P.

5.5.3 Characteristics of Symmetric and Antisymmetric Tensors

A few characteristics of symmetric and antisymmetric tensors, and of the operation of symmetrization and antisymmetrization of tensors, are in order:

1. Consider a tensor $T_{\alpha_1\alpha_2\ldots\alpha_p}$ of rank p (say).
   - If it is completely symmetric then one can express $T_{\alpha_1\alpha_2\ldots\alpha_p} = T_{(\alpha_1\alpha_2\ldots\alpha_p)}$.
   - If it is partially symmetric, say, in the first n of the p indices, then $T_{\alpha_1\alpha_2\ldots\alpha_p} = T_{(\alpha_1\alpha_2\ldots\alpha_n)\alpha_{n+1}\ldots\alpha_p}$.
   - If it is completely antisymmetric then one can express $T_{\alpha_1\alpha_2\ldots\alpha_p} = T_{[\alpha_1\alpha_2\ldots\alpha_p]}$.
   - If it is partially antisymmetric, say, in the first n of the p indices, then $T_{\alpha_1\alpha_2\ldots\alpha_p} = T_{[\alpha_1\alpha_2\ldots\alpha_n]\alpha_{n+1}\ldots\alpha_p}$.

2. The scalar product of a symmetric tensor and an antisymmetric tensor vanishes identically. Consider for e.g. two second rank tensors $S_{\alpha\beta} = S_{(\alpha\beta)}$ and $A_{\mu\nu} = A_{[\mu\nu]}$. Their scalar product

$$\begin{aligned} S_{\alpha\beta}A^{\alpha\beta} &= S_{\beta\alpha}A^{\beta\alpha} \quad : \text{ by interchanging the dummy indices } \alpha,\beta \\ &= S_{\alpha\beta}\left[-A^{\alpha\beta}\right] \quad : \text{ by using the symmetry and antisymmetry of } S_{\alpha\beta} \text{ and } A^{\alpha\beta} \\ &= 0\,. \end{aligned} \tag{5.54}$$

   In fact, the contracted product any two tensors of rank ≥ 2 vanishes identically, if one tensor is symmetric and the other tensor is antisymmetic in the pair of indices which are being contracted. This is irrespective of whether the tensors are partially or completely antisymmetric, whether they are of equal rank, and whether the contracted product involves more than two identical contra and covariant indices. For e.g.

$$S_{\alpha(\beta\gamma)}A^{[\beta\gamma]\delta} = 0\,, \quad S_{(\alpha\beta)}A^{[\alpha\beta\gamma\delta]} = 0\,, \quad S_{(\alpha\beta\gamma\delta)}A_{\mu}{}^{[\beta\delta]}{}_{\nu} = 0\,, \quad S^{\lambda}{}_{(\alpha|\beta|\gamma)\delta}A^{[\alpha|\mu\delta|\gamma]} = 0\,, \quad \cdots\,. \tag{5.55}$$

3. In D-dimensions, the number of independent components of a rank p tensor $T_{\alpha_1\ldots\alpha_p}$, and of the tensors $T_{(\alpha_1\ldots\alpha_p)}$ and $T_{[\alpha_1\ldots\alpha_p]}$ obtained by its symmetrization and antisymmetrization, are respectively

$$\mathcal{N} = D^p\,, \quad \mathcal{N}_{sym} = \binom{D+p-1}{p} = \frac{(D+p-1)!}{p!\,(D-1)!}\,, \quad \mathcal{N}_{antisym} = \binom{D}{p} = \frac{D!}{p!\,(D-p)!}\,. \tag{5.56}$$

   For e.g. if $p = 2$, then $T_{\alpha_1\ldots\alpha_p} = T_{\alpha_1\alpha_2}$ has $\mathcal{N} = D^2$ components, and the number of independent components of the symmetric tensor $T_{(\alpha_1\alpha_2)}$ could be computed as

$$\mathcal{N}_{sym}\Big|_{p=2} = \mathcal{N}(\text{diag}) + \frac{1}{2}\mathcal{N}(\text{off-diag}) = D + \frac{(D^2-D)}{2} = \frac{D\,(D+1)}{2}\,, \tag{5.57}$$

   which matches with that obtained from the expression for $\mathcal{N}_{sym}$ in Eq. (5.56), viz.

$$\frac{(D+2-1)!}{2!\,(D-1)!} = \frac{D(D+1)}{2}\,.$$

   Similarly, keeping in mind that no diagonal elements exist for the antisymmetric tensor $T_{[\alpha_1\alpha_2]}$, the number of its independent components can be computed as

$$\mathcal{N}_{antisym}\Big|_{p=2} = \frac{1}{2}\mathcal{N}(\text{off-diag}) = \frac{(D^2-D)}{2} = \frac{D\,(D-1)}{2}\,, \tag{5.58}$$

   which matches with that one gets using the expression for $\mathcal{N}_{antisym}$ in Eq. (5.56). Moreover, since

$$\frac{D(D+1)}{2} + \frac{D(D-1)}{2} = D^2\,,$$

   the tensors $T_{(\alpha_1\alpha_2)}$ and $T_{[\alpha_1\alpha_2]}$, due to symmetrization and antisymmetrization, add up to give the original tensor $T_{\alpha_1\alpha_2}$. This is a unique feature of rank-2 tensors in a general D-dimensional spacetime.

4. The above expression for $\mathcal{N}_{antisym}$ in Eq. (5.56) is also suggestive from the point of view of its validity only for tensors of rank $p \leq D$. This is evident from the reduced expression:

$$\mathcal{N}_{antisym} = \frac{D!}{p!\,(D-p)!} = \frac{D\,(D-1)\,(D-2)\ldots(D-p+1)}{p!}\,, \tag{5.59}$$

which vanishes whenever $p = D+1$. Therefore, for any tensor whose rank exceeds the dimensionality, antisymmetrization identically leads to zero. For e.g. in 3-dimensions ($D = 3$) the maximum rank of a completely antisymmetric tensor is 3 (i.e. the tensor is $T_{[ijk]}$), and the tensor has *only one* non-vanishing independent component T_{123}. The other non-vanishing components are all equal to T_{123} in magnitude (e.g. T_{231}, $|T_{312}|$). Similarly, in (3+1)-dimensions ($D = 4$) the maximum rank of a completely antisymmetric tensor is 4 (i.e. the tensor is $T_{[\mu\nu\lambda\sigma]}$), and the only non-vanishing independent component of the tensor is T_{0123}. In fact, complete antisymmetry implies all the indices of a given tensor must be different. So when the rank equals the dimensionality, i.e. the number of tensor indices equals the number of dimensions, there is only one choice left for all the tensor indices to be unequal.

5.5.4 The Levi-Civita Tensor and the Generalized Kronecker Delta

In D-dimensions, the maximum ranked completely antisymmetric tensor is proportional to the *Levi-Civita tensor* defined by*

$$T_{[\alpha_1\alpha_2\ldots\alpha_D]} \propto \epsilon_{\alpha_1\alpha_2\ldots\alpha_D} = \begin{cases} -1 & \text{for } \alpha_1,\alpha_2,\ldots,\alpha_D \text{ an even permutation of } 0,1,\ldots,D \\ +1 & \text{for } \alpha_1,\alpha_2,\ldots,\alpha_D \text{ an odd permutation of } 0,1,\ldots,D \\ 0 & \text{otherwise}\,. \end{cases} \tag{5.60}$$

$$T^{[\alpha_1\alpha_2\ldots\alpha_D]} \propto \epsilon^{\alpha_1\alpha_2\ldots\alpha_D} = \begin{cases} +1 & \text{for } \alpha_1,\alpha_2,\ldots,\alpha_D \text{ an even permutation of } 0,1,\ldots,D \\ -1 & \text{for } \alpha_1,\alpha_2,\ldots,\alpha_D \text{ an odd permutation of } 0,1,\ldots,D \\ 0 & \text{otherwise}\,. \end{cases} \tag{5.61}$$

The following contraction identities exist between $\epsilon^{\mu\nu\lambda\sigma}$ and $\epsilon_{\mu\nu\lambda\sigma}$ in $(3+1)$-dimensions:

$$\epsilon^{\mu\nu\lambda\sigma}\epsilon_{\mu\nu\lambda\sigma} = -4!\,, \quad \epsilon^{\mu\nu\lambda\sigma}\epsilon_{\alpha\nu\lambda\sigma} = -3!\,\delta^{\mu}{}_{\alpha}\,, \quad \epsilon^{\mu\nu\lambda\sigma}\epsilon_{\alpha\beta\lambda\sigma} = -2!\,\delta^{\mu\nu}{}_{\alpha\beta}\,, \quad \epsilon^{\mu\nu\lambda\sigma}\epsilon_{\alpha\beta\gamma\sigma} = -\delta^{\mu\nu\lambda}{}_{\alpha\beta\gamma}\,, \tag{5.62}$$

where

$$\delta^{\mu\nu}{}_{\alpha\beta} = \begin{vmatrix} \delta^{\mu}{}_{\alpha} & \delta^{\nu}{}_{\alpha} \\ \delta^{\mu}{}_{\beta} & \delta^{\nu}{}_{\beta} \end{vmatrix}\,, \quad \delta^{\mu\nu\lambda}{}_{\alpha\beta\gamma} = \begin{vmatrix} \delta^{\mu}{}_{\alpha} & \delta^{\nu}{}_{\alpha} & \delta^{\lambda}{}_{\alpha} \\ \delta^{\mu}{}_{\beta} & \delta^{\nu}{}_{\beta} & \delta^{\lambda}{}_{\beta} \\ \delta^{\mu}{}_{\gamma} & \delta^{\nu}{}_{\gamma} & \delta^{\lambda}{}_{\gamma} \end{vmatrix}\,, \quad \ldots\,, \tag{5.63}$$

are the *Generalized Kronecker Delta* tensors.

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*Most appropriately, the definitions hold for the covariant and contravariant Levi-Civita tensor densities $\varepsilon_{\alpha_1\alpha_2\ldots\alpha_D}$ and $\varepsilon^{\alpha_1\alpha_2\ldots\alpha_D}$. Tensor densities are the most general entities which follow linear, homogeneous laws of transformation under the transformation of coordinates, and tensors form only a sub-class of tensor densities. One can distinguish a tensor from a tensor density considering that the latter involves in its transformation law an extra factor depending on a certain integral power W (called the *weight*) of the Jacobian $|\partial x/\partial x'|$:

$$\mathcal{T}'^{\mu\cdots}{}_{\nu\cdots} = \left|\frac{\partial x}{\partial x'}\right|^{W} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \cdots \frac{\partial x^{\beta}}{\partial x'^{\nu}} \cdots \mathcal{T}^{\alpha\cdots}{}_{\beta\cdots}\,.$$

However, since the Jacobian is identically equal to unity for the Lorentz transformations in Special Relativity, the above transformation law for the tensor density $\mathcal{T}^{\alpha\cdots}{}_{\beta\cdots}$ reduces to the transformation law for a tensor $T^{\alpha\cdots}{}_{\beta\cdots}$:

$$T'^{\mu\cdots}{}_{\nu\cdots} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \cdots \frac{\partial x^{\beta}}{\partial x'^{\nu}} \cdots T^{\alpha\cdots}{}_{\beta\cdots}\,.$$

Accordingly, there is no distinction between the Levi-Civita tensor density $\varepsilon_{\alpha_1\alpha_2\ldots\alpha_D}$ of weight $W = -1$ and the tensor $\epsilon_{\alpha_1\alpha_2\ldots\alpha_D}$, and also between the Levi-Civita tensor density $\varepsilon^{\alpha_1\alpha_2\cdots\alpha_D}$ of weight $W = +1$ and the tensor $\epsilon^{\alpha_1\alpha_2\cdots\alpha_D}$, in Special Relativity.

5.5.5 Tensor Duality

In D-dimensions the total number of independent components of a completely antisymmetric tensor of rank p is the same as that of a completely antisymmetric tensor of rank $(D-p)$, since

$$\mathcal{N}_{antisym}(D,p) \;=\; \frac{D!}{p!\,(D-p)!} \;=\; \frac{D!}{(D-p)!\,[D-(D-p)]!} \;=\; \mathcal{N}_{antisym}(D,D-p)\;. \tag{5.64}$$

This implies that there exists a *duality* between a rank p completely antisymmetric tensor and a rank $(D-p)$ completely antisymmetric tensor in D-dimensions. Such a duality relationship is given by

$$A_{\alpha_1\alpha_2\ldots\alpha_p} \;=\; A_{[\alpha_1\alpha_2\ldots\alpha_p]} \;=\; p!\;\epsilon_{\alpha_1\alpha_2\ldots\alpha_p\alpha_{p+1}\ldots\alpha_D}\;{}^\star A^{\alpha_{p+1}\ldots\alpha_D}\;, \tag{5.65}$$

where ${}^\star A^{\alpha_{p+1}\ldots\alpha_D}$ is called the (*Hodge*-)dual of $A_{\alpha_1\alpha_2\ldots\alpha_p}$. The inverse duality relation is given by

$${}^\star A_{\alpha_1\alpha_2\ldots\alpha_p} \;=\; {}^\star A_{[\alpha_1\alpha_2\ldots\alpha_p]} \;=\; \frac{1}{p!}\,\epsilon_{\alpha_1\alpha_2\ldots\alpha_p\alpha_{p+1}\ldots\alpha_D}\;A^{\alpha_{p+1}\ldots\alpha_D}\;. \tag{5.66}$$

The indices of $A_{\alpha\beta\ldots}$ (or $A^{\alpha\beta\cdots}$) and ${}^\star A_{\alpha\beta\ldots}$ (or ${}^\star A^{\alpha\beta\cdots}$) are raised (or lowered) in the usual way using the metric tensor. Hence, in 3-dimensions:

$$\begin{aligned}
A_{ijk} = 3!\;\epsilon_{ijk}\;{}^\star A \;&\Longrightarrow\; A^{ijk} = 3!\;\epsilon^{ijk}\;{}^\star A\;, &\qquad {}^\star A_{ijk} = \frac{1}{3!}\;\epsilon_{ijk}\;A \;&\Longrightarrow\; {}^\star A^{ijk} = \frac{1}{3!}\;\epsilon^{ijk}\;A\;,\\
A_{ij} = 2!\;\epsilon_{ijk}\;{}^\star A^k \;&\Longrightarrow\; A^{ij} = 2!\;\epsilon^{ijk}\;{}^\star A_k\;, & {}^\star A_{ij} = \frac{1}{2!}\;\epsilon_{ijk}\;A^k \;&\Longrightarrow\; {}^\star A^{ij} = \frac{1}{2!}\;\epsilon^{ijk}\;A_k\;,\\
A_i = \epsilon_{ijk}\;{}^\star A^{jk} \;&\Longrightarrow\; A^i = \epsilon^{ijk}\;{}^\star A_{jk}\;, & {}^\star A_i = \epsilon_{ijk}\;A^{jk} \;&\Longrightarrow\; {}^\star A^i = \epsilon^{ijk}\;A_{jk}\;,\\
A = \epsilon_{ijk}\;{}^\star A^{ijk} &= \epsilon^{ijk}\;{}^\star A_{ijk}\;, & {}^\star A = \epsilon_{ijk}\;A^{ijk} &= \epsilon^{ijk}\;A_{ijk}\;,
\end{aligned} \tag{5.67}$$

whereas in $(3+1)$-dimensions:

$$\begin{aligned}
A_{\alpha\beta\gamma\delta} = 4!\epsilon_{\alpha\beta\gamma\delta}\;{}^\star A \;&\Longrightarrow\; A^{\alpha\beta\gamma\delta} = 4!\epsilon^{\alpha\beta\gamma\delta}\;{}^\star A\,, &\qquad {}^\star A_{\alpha\beta\gamma\delta} = \frac{1}{4!}\epsilon_{\alpha\beta\gamma\delta}\;A \;&\Longrightarrow\; {}^\star A^{\alpha\beta\gamma\delta} = \frac{1}{4!}\epsilon^{\alpha\beta\gamma\delta}\;A\,,\\
A_{\alpha\beta\gamma} = 3!\epsilon_{\alpha\beta\gamma\delta}\;{}^\star A^\delta \;&\Longrightarrow\; A^{\alpha\beta\gamma} = 3!\epsilon^{\alpha\beta\gamma\delta}\;{}^\star A_\delta\,, & {}^\star A_{\alpha\beta\gamma} = \frac{1}{3!}\epsilon_{\alpha\beta\gamma\delta}\;A^\delta \;&\Longrightarrow\; {}^\star A^{\alpha\beta\gamma} = \frac{1}{3!}\epsilon^{\alpha\beta\gamma\delta}\;A_\delta\,,\\
A_{\alpha\beta} = 2!\epsilon_{\alpha\beta\gamma\delta}\;{}^\star A^{\gamma\delta} \;&\Longrightarrow\; A^{\alpha\beta} = 2!\epsilon^{\alpha\beta\gamma\delta}\;{}^\star A_{\gamma\delta}\,, & {}^\star A_{\alpha\beta} = \frac{1}{2!}\epsilon_{\alpha\beta\gamma\delta}\;A^{\gamma\delta} \;&\Longrightarrow\; {}^\star A^{\alpha\beta} = \frac{1}{2!}\epsilon^{\alpha\beta\gamma\delta}\;A_{\gamma\delta}\,,\\
A_\alpha = \epsilon_{\alpha\beta\gamma\delta}\;{}^\star A^{\beta\gamma\delta} \;&\Longrightarrow\; A^\alpha = \epsilon^{\alpha\beta\gamma\delta}\;{}^\star A_{\beta\gamma\delta}\,, & {}^\star A_\alpha = \epsilon_{\alpha\beta\gamma\delta}\;A^{\beta\gamma\delta} \;&\Longrightarrow\; {}^\star A^\alpha = \epsilon^{\alpha\beta\gamma\delta}\;A_{\beta\gamma\delta}\,,\\
A = \epsilon_{\alpha\beta\gamma\delta}\;{}^\star A^{\alpha\beta\gamma\delta} &= \epsilon^{\alpha\beta\gamma\delta}\;{}^\star A_{\alpha\beta\gamma\delta}\,, & {}^\star A = \epsilon_{\alpha\beta\gamma\delta}\;A^{\alpha\beta\gamma\delta} &= \epsilon^{\alpha\beta\gamma\delta}\;A_{\alpha\beta\gamma\delta}\,.
\end{aligned} \tag{5.68}$$

5.6 Tensor Calculus

5.6.1 Tensor Differentiation: Grad, Div, Curl and D'Alembertian operators

Differentiation of tensors in a $(3+1)$-dimensional space-time is analogous to that in the ordinary Euclidean 3-dimensional space. We consider only the special relativistic scenario, i.e. the $(3+1)$-dimensional Minkowski space-time, in which the the differential operators and operations are as follows:

Covariant and Contravariant Differential operators:

$$\boxed{\partial_\mu \equiv \frac{\partial}{\partial x^\mu} \;=\left(\frac{\partial}{\partial x^0},\frac{\partial}{\partial x^i}\right) \;\equiv\; (\partial_0,\nabla)} \quad\text{and}\quad \boxed{\partial^\mu \equiv \frac{\partial}{\partial x_\mu} \;=\left(\frac{\partial}{\partial x_0},\frac{\partial}{\partial x_i}\right) \;\equiv\; (\partial^0,-\nabla)}\,, \tag{5.69}$$

where we have used the notations:

$$x_0 = \eta_{0\beta}\,x^\beta = \eta_{00}\,x^0 = x^0 \equiv ct \;\Longrightarrow\; \partial^0 \equiv \frac{\partial}{\partial x_0} = \frac{\partial}{\partial x^0} \equiv \partial_0 = \frac{1}{c}\frac{\partial}{\partial t}\;, \tag{5.70}$$

$$x_i = \eta_{i\beta}\,x^\beta = \eta_{ij}\,x^j = -\delta_{ij}\,x^j = -x^i \equiv -(\vec{\boldsymbol{x}})^i \;\Longrightarrow\; \partial^i \equiv \frac{\partial}{\partial x_i} = -\frac{\partial}{\partial x^i} \equiv -\partial_i = -(\nabla)^i\;, \tag{5.71}$$

∇ being the ordinary three-gradient operator.

Note that the above co(contra)variant operators [Eq. (5.69)] have component structures just opposite to the co(contra)variant four-vectors, viz. $A_\mu = (A_0, -\vec{\boldsymbol{A}})$, and $A^\mu = (A^0, \vec{\boldsymbol{A}})$, where $\vec{\boldsymbol{A}}$ is the ordinary three-vector.

Covariant Gradient: In a $(3+1)$-dimensional space-time, the *covariant gradient* (or, the *four-gradient*) of a four-scalar $\phi(t, \vec{\boldsymbol{x}})$, defined by

$$\boxed{\partial_\mu \phi \;\equiv\; \frac{\partial \phi}{\partial x^\mu} \;=\; \left(\frac{\partial \phi}{\partial x^0}, \nabla\phi\right) = \left(\frac{1}{c}\frac{\partial \phi}{\partial t}, \nabla\phi\right)}, \tag{5.72}$$

gives a covariant vector. A similar operation, sometimes called the *contravariant gradient*, on the four-scalar $\phi(t, \vec{\boldsymbol{x}})$, yields a contravariant vector

$$\partial^\mu \phi \;=\; g^{\mu\nu}\,\partial_\nu \phi \;\equiv\; \frac{\partial \phi}{\partial x_\mu} \;=\left(\frac{1}{c}\frac{\partial \phi}{\partial t}, -\nabla\phi\right). \tag{5.73}$$

Covariant Divergence: In a (3+1)-dimensional space-time, the *covariant divergence* (or, the *four-divergence*) of a (contravariant) four-vector $A^\mu(t, \vec{\boldsymbol{x}})$, i.e. a contravariant tensor of rank 1, defined by

$$\boxed{\partial_\mu A^\mu \;\equiv\; \frac{\partial A^\mu}{\partial x^\mu} \;=\; \frac{\partial A^0}{\partial x^0} \;+\; \frac{\partial A^i}{\partial x^i} \;=\; \frac{1}{c}\frac{\partial A^0}{\partial t} \;+\; \nabla\cdot\vec{\boldsymbol{A}}}, \tag{5.74}$$

gives a four-scalar, i.e. a tensor of rank 0. Similarly, the (covariant) divergence of a contravariant tensor $T^{\mu\nu}$ of rank 2, defined by

$$\boxed{\partial_\mu T^{\mu\nu} \;\equiv\; \frac{\partial T^{\mu\nu}}{\partial x^\mu} \;=\; \frac{\partial T^{0\nu}}{\partial x^0} \;+\; \frac{\partial T^{i\nu}}{\partial x^i} \;=\Big(\left[\partial_0 T^{00} + \partial_i T^{i0}\right], \left[\partial_0 T^{0j} + \partial_i T^{ij}\right]\Big)}, \tag{5.75}$$

yields a (contravariant) four-vector, i.e. a contravariant tensor of rank 1.

In general, we thus have for any contravariant tensor of rank m, the operation of (covariant) divergence giving a contravariant tensor of rank $(m-1)$. An associated operation, sometimes called the *contravariant divergence* is similarly defined, e.g.

$$\partial_\mu A^\mu \;=\; \partial^\mu A_\mu\,, \qquad \partial_\mu T^{\mu\nu} \;=\; \partial^\mu T_\mu^{\ \nu}\,, \qquad \partial_\mu T^{\mu\nu\lambda} \;=\; \partial^\mu T_\mu^{\ \nu\lambda}\,, \quad \ldots\,. \tag{5.76}$$

Covariant Curl: In a $(3+1)$-dimensional space-time, the *covariant curl* (or, the *four-curl*) of a covariant vector $A_\nu(t, \vec{\boldsymbol{x}})$, defined by

$$\boxed{C^{\alpha\beta} \;=\; C^{[\alpha\beta]} \;=\; \frac{1}{2}\epsilon^{\alpha\beta\mu\nu}\left(\partial_\mu A_\nu - \partial_\nu A_\mu\right) = \;\epsilon^{\alpha\beta\mu\nu}\,\partial_\mu A_\nu}. \tag{5.77}$$

gives a contravariant antisymmetric tensor of rank 2.

Note that this generalizes the ordinary (Euclidean) curl of a three-vector $A_k(\vec{\boldsymbol{x}})$, which is itself a three-vector with the i-th component given as $(\partial_j A_k - \partial_k A_j)$, i.e.

$$\left(\nabla\times\vec{\boldsymbol{A}}\right)^i \;=\; \frac{1}{2}\epsilon^{ijk}\left(\partial_j A_k - \partial_k A_j\right) = \;\epsilon^{ijk}\,\partial_j A_k\,. \tag{5.78}$$

The D'Alembertian Operator: In a $(3+1)$-dimensional space-time, there is another useful operator, known as the *D'Alembertian operator*, defined as

$$\boxed{\Box^2 \;\equiv\; \partial_\alpha \partial^\alpha \;=\; \partial_0 \partial^0 \;+\; \partial_i \partial^i \;=\; \frac{1}{c^2}\frac{d^2}{dt^2} \;-\; \nabla^2}. \tag{5.79}$$

This is the generalized version of the 3-dimensional Laplacian operator $(\nabla^2 = \nabla\cdot\nabla)$, and is primarily used in the covariant formulation of wave equations.

5.6.2 The Four-volume and Tensor Integration

The invariant Four-volume element: The volume element in $(3+1)$-dimensions is given by

$$d\mathcal{V} \; = \; d^4x \; \equiv \; dt\, d^3x \; . \tag{5.80}$$

Under a coordinate transformation $x^\mu \longrightarrow x'^\mu = x'^\mu(x^0, x^1, x^2, x^3)$ [*cf.* Eq. (5.1)], the volume element transforms to

$$d\mathcal{V}' \; = \; d^4x' \; = \left|\frac{\partial x'}{\partial x}\right| d^4x \; = \left|\frac{\partial x'}{\partial x}\right| d\mathcal{V} \; , \tag{5.81}$$

i.e. d^4x is actually a scalar density (or, a tensor density of rank 0). Taking now the determinant of both sides of the transformation equation of the metric tensor, viz.

$$g'_{\mu\nu} \; = \; \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\, g_{\alpha\beta} \qquad [\textit{cf.}\ \text{Eq. } (5.31)] \, ,$$

we have

$$g' \equiv \det\left(g'_{\mu\nu}\right) = \left|\frac{\partial x}{\partial x'}\right|^2 \det\left(g_{\alpha\beta}\right) \equiv \left|\frac{\partial x}{\partial x'}\right|^2 g \; . \tag{5.82}$$

Since the determinant g of the metric tensor is negative in spaces having the Lorentzian signature $(+\,,-\,,-\,,-)$, therefore in such spaces

$$\sqrt{-g'} = \left|\frac{\partial x}{\partial x'}\right| \sqrt{-g} \; . \tag{5.83}$$

Substituting this in Eq. (5.81), and recalling that

$$\left|\frac{\partial x'}{\partial x}\right| \left|\frac{\partial x}{\partial x'}\right| \; = \; \mathcal{J}(x \to x')\mathcal{J}(x' \to x) \; = \; 1 \qquad [\textit{cf.}\ \text{Eq. } (5.6)] \, ,$$

we find

$$\sqrt{-g'}\, d\mathcal{V}' \; = \; \sqrt{-g}\, d\mathcal{V} \; , \tag{5.84}$$

i.e. the *invariant four-volume element* is a scalar (or, a tensor of rank 0) given by

$$\boxed{\sqrt{-g}\, d\mathcal{V} \; = \; \sqrt{-g}\, d^4x \; \equiv \; \sqrt{-g}\, dt\, d^3x} \; . \tag{5.85}$$

In Special Relativity, we have $g_{\alpha\beta}$ identified with the Minkowski metric tensor $\eta_{\alpha\beta}$, whence $g = \det\left(\eta_{\alpha\beta}\right) = -1$, and the invariant four-volume element is simply $d^4x = dt\, d^3x$ — a Lorentz scalar.

Tensor Integration: In principle, integration of tensors over the invariant four-volume yield tensor-like quantities. One defines surface and volume integrals by defining infinitesimal surface and volume elements in a $(3+1)$-dimensional space-time, just as in the ordinary 3-dimensional Euclidean space. One also has the generalizations of the Euclidean-space *Divergence theorem*, *Stokes theorem* etc. in a $(3+1)$-dimensional space-time. However, we skip for brevity discussing them in these lectures.